

GAHAN SARAIYA

☎ +91 7359508173 ✉ gahansaraiya@gmail.com 💼 [linkedin.com/in/gahan-saraiya](https://www.linkedin.com/in/gahan-saraiya) 🐙 github.com/gahan9

OBJECTIVE

With over 8+ years of experience as a Senior Software Engineer, currently working at AMD India Pvt Ltd. for managing AI/ML workload execution on HPC device (GPU). I possess a strong command over Python, HPC, UEFI Firmware, and DevOps, complemented by a thorough understanding of their workflows and architectures. My enthusiasm for open-source contribution and knowledge dissemination reflects my commitment to being a philanthropist in the realm of technology. I am eager to channel my expertise into the architectural design of optimized quantum machine learning algorithms tailored for multi-programming environments, as well as to explore the synergies between these cutting-edge technologies and DevOps practices.

SOFTWARE TOOLS

Languages	Python, C, C++, Shell Scripting, SQL, LaTeX, YAML
Frameworks	Tensorflow, Pytorch, LangGraph, FAISS, TensorRT, Pytest, Edk2, PyCUDA, Pandas, UEFI, Selenium, Redfish
Architecture Design Tools and Services	Parallel Programming System Architecture, ROCm, HIP Programming GitHub Actions, Claude, Cursor, LLM APIs, Restful APIs, Teamcity, Terraform, Docker, Docker Compose, Jenkins, GiT
Cloud Technologies	GCP, AWS, Azure, Vertex AI, Digital Ocean, JFrog Artifactory, TeamCity, JIRA, AWS Sage Maker
System Level IP	RAS, Compute Partitioning, Boot time optimization, CPER

EXPERIENCE

Senior System Software Designer at AMD India Pvt Ltd Mar 2024 - Present

- Conducting functional and performance benchmarking of HPC devices (GPUs) by developing and executing comprehensive test suites aligned with industry standards.
- Develop Load balanced optimal GPU performance criteria with thermal limits, power consumption, memory usage, and core utilization.
- Incorporating state-of-the-art industry standards design models, implementing and utilizing frameworks like TensorFlow, PyTorch, and GEMM operations.
- Provided precise steps for RAS (Reliability, Availability, and Serviceability) injection on MI2XX and MI3XX platforms.
- Designed precise steps for range of firmware capabilities, including partitioning, System Management Interface (SMI), HIP profiling, and parallel execution.
- Collaborating with a diverse team of engineers and researchers.

Senior Firmware Development Engineer at Intel Technology India Pvt Ltd Oct 2023 - Mar 2024

- Platform provisioning and infrastructure setup for mass automation of HPC products.
- Platform Orchestration Layer design of Universal Scalable Firmware (USF).
- Harnessing Azure OpenAI Generative AI adaptation for Firmware Development Productivity for assessment code review, guidance, basic security checks of code.
- GitHub Action and JIRA integration for managing Continuous deployment and test through CloudBees Jenkins with baremetal and docker nodes.
- Enhancing firmware-based boot-time performance for notebook devices.

Firmware Development Engineer at Intel Technology India Pvt Ltd Jun 2020 - Sept 2023

- Lead the next generation client platform and architecture design at USF POL alongside the domain ownership for Automation Framework, Platform Provisioning, Applied Generative AI & Prompt Engineering and Quantum Firmware Design.
- The end-to-end Software Development Life Cycle is managed, which includes licensing, legals, and IP planning, as well as applicable Code Security and Static Code Analysis.
- Jenkins configuration for Security tools - Snyk, Coverity
- Improving github code review hooks for python packaging and linting through bandit and flask as Jeninks pipeline
- automated testing of firmware functional validation of Firmware with TeamCity.
- Mentoring and knowledge development engagement activities.

Firmware Engineer Intern at Intel Technology India Pvt Ltd

Jun 2019 - May 2020

- BIOS Firmware Development for Client products with the framework EDK2, UEFI Firmware, Involvement with assessing new debugging solutions and OpenBoard Package

Teaching Assistantship at Nirma University

Aug 2018 - May 2019

- Data Mining and Visulaization

Software Developer at Quixom Technology

Jul 2017 - Apr 2018

- SaaS-Employee management system (chat, surveys, benchmark, analytics...) and Restful API Development
- Socket programming in python to receive continuous data from IoT device and display chunks of data meaningfully
- Streaming service with Web-Portal, DRF and Kodi across platforms

Freelance Full-stack Developer

2015 - Apr 2019

- Cloud solution for Data gathering and webscraping with selenium and GCP.
- Designing template and banners for High resolution prints
- Inventory management system for Book store with Django.

PROJECTS

Agentic AI for Software Engineering Modernization

2025 - Present

Description: Modernized enterprise workflows, repository management, and code review processes using multi-tool compatible Agentic AI and GenAI architectures (Claude, Cursor).

- Developed robust tools, automated git hooks, and modular agents; implemented security compliance and token spend restrictions.

Context-Aware Task Routing & Triage with LangGraph

2025 - Present

Description: Led team sprint goals to achieve automated context-aware task/issue routing and resolution using graph-based LLMs.

- Designed smart routing using LangGraph and semantic FAISS indexes, integrating OpenAI, Vertex AI, and local LLM APIs.
- Built connectors for GitHub, Jira, and historical systems, with a persona counselor model for summarizing and adapting findings.

DevOps Management for Cluster Validation

2025 - Present

Description: Engineered CI pipeline configurations for validation of heterogeneous computing systems and log collection.

- Developed automated telemetry log shipping to S3 buckets with integrated cloud network security controls and compliance.

GPU Performance Analysis for Data Center Workloads

2024 - Present

Description: Solution design for in-band profiling through System Management Interface

- **Architected In-Band Profiling Solutions:** Designed and implemented robust in-band profiling mechanisms utilizing the System Management Interface (SMI) for comprehensive GPU telemetry.
- **Dynamic Workload Orchestration:** Developed systems for dynamic workload creation, intelligently adapting to real-time hardware resource availability (e.g., GPU utilization, memory) to optimize testing and analysis.
- **Comprehensive Hardware Metric Evaluation:** Engineered processes for capturing and analyzing critical hardware metrics, including GPU temperature, power consumption, and resource utilization, directly through SMI and in-band channels.
- **Granular Program Performance Analysis:** Implemented detailed program profiling techniques, including the capture and analysis of kernel call stack timings, to pinpoint performance bottlenecks and inefficiencies.
- **Advanced Analytical Metric Design:** Innovated a methodology to map multiple disparate key factors (e.g., thermal, power, utilization, kernel timings) into a unified, insightful data metric, significantly enhancing the depth and clarity of system performance analysis.

Prompt Engineering and Generative AI for firmware stack

2023 - 2024

Description: Solution design for accelerated silicon tape-in with accelerating firmware deliveries

- Work task summarization with Azure LLM Open AI services (GPT3.5 and GPT4)
- Code Review against standard static code analysis and common vulnerabilities finding
- Generation of Code Enhancement suggestion

Platform Provisioning with Pre-Production Reference Board

2023 - 2024

Description: Platform Provisioning with automating most common features for pre-production server processor such as:

- Remote BMC and BIOS Firmware Flashing
- OS Deployment from artifactory through Jenkins Pipeline
- Boot to EFI Shell and OS
- Configuration of Initial system setup with Jenkins pipeline steps with Baseboard management controller RedFish API

Universal Scalable Firmware

2022 - 2024

Description: Architecture design of Platform orchestration layer for Firmware Configuration

- Motivation of workgroup - [Firmware Configuration – Past, Present, and Future](#)
- [Firmware Configuration](#)

Firmware Automation Framework

2020 - Present

Description: Lead SDL for an automation foundation framework for high scaled volume of silicon devices with edk2 UEFI firmware;

- Seamless configuration of system without any manual intervention for BIOS configuration.
- SDL activities like legal compliance, security, licensing, static code analysis
- Continuous deployment and test with jenkins and tox configurations
- Release Artifact management to JFrog Artifactory using GitHub Actions

- <https://github.com/intel/xml-cli>

Multiview Video Summarization

2019 - 2020

Description: Summarizing video of multiple view angles of egocentric location in to one to generate effective video summary, minimizing overhead to explore multiple cam feed to discover accidents or anomalies.

EDUCATION

M.Tech Computer Science

Jul 2018 - Jun 2020

Nirma University, GPA: 8.49

B.E. in Computer Science and Engineering

Aug 2013 - Apr 2017 Gujarat Technological University,

CGPA: 8.08

12th Science, GSHSEB, Class XII

Jun 2011 - Mar 2013

Science Stream, 74.92%

CERTIFICATIONS

2025 [AMD ROCm Star - Application Developer](#) by AMD Adaptive Computing Training Team

2025 [On-Device AI](#) by Qualcomm & DeepLearning.ai

2025 [Essential Terraform in AWS](#)

2021 [Product Assurance and Security Yellow Belt - Software](#)

2020 Natural Language Processing with Classification and Vector Spaces

[Coursera S9DGGMCDMBGQ](#)

2020 Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

[Coursera SMWQJKJYJW6G](#)

2020 Structuring Machine Learning Projects - [Coursera SEJML7H3TN5Z](#)

2020 Neural Networks and Deep Learning - [Coursera S2TETZHTLUSL](#)

2020 Machine Learning - [Coursera XVNXNPPZ3JMA](#)

ACHIEVEMENTS

2026 Top 10 in AI Hackathon: Hack2Skill Prompt Wars by Google (In-person, Bangalore)

2026 Rank #3 in Virtual Prompt Wars Hackathon

2023 Firmware Configuration Presentation at UEFI Developers Conference & Plugfest

2023 Tech Talk on Harnessing Multiprogramming across CPU, GPU and QPU (Quantum Processing Unit)

2021 Expert Lecture on CUDA Programming

2019-Present 4000+ reputation on Stackoverflow

2017-Present Active volunteer on Peer Learning, Cultural and team building activity

2018 Rank #1 for Python challenges on hackerrank

2018 Rank #112 from 2 lacs+ participants in Techgig's National Coding Contest 2018

PUBLICATIONS

- [HIP Parallel Programming with ROCm](#) Lecture Series, YouTube 2025
- [Beyond the Power — Data Center & BMC](#) Lecture Series, YouTube 2025
- [System Design for Parallel Programming](#) Lecture Series, YouTube 2025
- [Firmware Configuration – Past, Present, and Future.](#) Presentation, UEFI DevCon 2023 2023